Writing to files in Python

CMSC 104 Section 2
November 9, 2026

Administrative notes

Project 3 will be assigned today

- It will be due in 2 weeks - before midnight on Monday, November 23

Quiz 3 is in class on Wednesday - November 11

- It covers all material up through last Wednesday
    - In other words, reading from files is covered, but I won't test you on writing to files at this point

Methods for writing to files

.write() - write one string to the file

.writelines() - write the elements of a list into a file.

Note: only strings can be written to a file. If you want to save int, float, or boolean variables to a file, you have to convert them to strings first.

This is different from printing to the screen, where you can directly print int’s or float’s or booleans

There is a “cheat code”

You can “print” to a file using the print statement itself

```
print(x, y, file=outfile)
```

But we won’t be using that in this class, because it’s actually fairly complicated

“Print” in this case does all the conversion into strings for you - but it might not be in the format you or another user may want

Examples of writing to a file

Can you read from and write to the same file at the same time?

Yes, but only if you’re careful.

Remember that when you open the file for writing, any previous data will be erased.

So what you should do is:

- Open the file for reading
- Read in the data
- Manipulate the data as needed
- Prepare the output
- Close the file
- Open the file for writing
- Write your new output

I/O to the same file

If what you want to do is read one line from a file, manipulate that data, write the modification back to the file, and then read the next line

- That’s not an easy thing to do with Python